Lesson Plan · Wednesday, November 11

Algebra I · Mr. Pandey · Module 3, Topic 1, Lesson 3: Modeling Linear Inequalities (Day 1 of 3) · essentials only (✪)

Objectives (students will…)

- Write and analyze a linear function from a context (rate of change, y-intercept).
- Use a graph to solve inequalities.
- Graph solution sets on a number line with open or closed circles.

TEKS

- **A.2C** write linear equations from a context
- **A.5B** solve linear inequalities in one variable
- Process: A.1A, A.1D

Key terms

- linear inequality

Materials

- Slides: m3t1-lesson3-modeling-linear-inequalities (.html/.pptx) · publisher slides with answers: BLMathematics_AlgI_M3_T1_L3_LessonSlides_ENG_with_answers.pptx
- Bell work: BELL WORK/T1-Bell-Work-Wed-11-11.pdf + key (online: m3t1-lesson3-bellwork-day1)
- Exit ticket: EXIT TICKET/T1-Exit-Ticket-Wed-11-11.pdf + key (written like the End of Topic 1 test)
- Video: videos/M3T1-Lesson3-Linear-Inequalities.mp4 · IEP and 504 versions of all of these: IEP/, Section 504/, and the -IEP / -504 PDFs
- Student Edition pp. 371–375 (Getting Started, Activity 3.1)
- Inequality-words chart (at least ≥, at most ≤, more than >, fewer than <)
- Rulers for the horizontal-line method
- Whiteboards

Bell work (on the board)

- Elijah starts with a $25 credit and earns $3.75 for each box he sells. Which equation gives his total sales, y, after selling x boxes? B. y = 3.75x + 25
- Which symbol means "at least"? H. ≥
- Which symbol means "no more than"? D. ≤
- What value of b makes 3.75b + 25 = 400 true? F. 100

Agenda (60 minutes)

Time	Activity	What to do / say
0–5	Bell work	Build the inequality-words chart on the board from Q1–3 (TE EB tip). Leave it up for all of Lesson 3.
5–20	✪ Getting Started: Fundraising Function · SE p. 372	Elijah: $25 credit + $3.75 per box. Q1 f(b) = 3.75b + 25. Q2 independent boxes sold, dependent total sales ($); slope 3.75, dollars per box; y-intercept 25, the starting credit.
20–45	✪ Activity 3.1: Modeling Linear Inequalities · SE pp. 373–375	Q1 goal $1600: horizontal line → 420 boxes. Define *linear inequality*. Q2 below the line: f(b) ≤ 1600, b ≤ 420, closed circle; only whole numbers of boxes make sense. Q3 above: b > 420, open circle. Q4 open = not included, closed = included. Q5 at least $925 b ≥ 240; less than $2050 b < 540; exactly $700 b = 180. **Test moves** (publisher slides 29–30, like test Q1, Q6): $40 + $15/week y = 15x + 40; 4 notebooks at $1.75 + x pens at $0.90 y = 0.90x + 7.
45–55	Exit ticket	1. Maya has $40 and earns $12 per hour. Which equation gives the total, y, she has after working x hours? y = 12x + 40 2. 4 notebooks at $1.75, pens at $0.90. Equation for x pens? y = 0.90x + 7 3. Sales of at most $1150: inequality for b? b ≤ 300, closed circle at 300, shaded left IEP/504: read aloud on request, properties/notes card, extra time.
55–60	Wrap up	Homework: Lesson 3 Assignment Practice 1a–d (SE p. 383). Tomorrow: solve inequalities with algebra.

Check for understanding

- Whiteboards: "Open or closed for x < 3?" open
- Point to the graph: "Where is f(b) > 1600?" above the line
- Activity 3.1 Q1–5 are the TE's check points.

After class

- Missed circles/shading → Skills Practice Set B (number lines).
- Missed writing the inequality → Skills Practice Set A.
- Re-teach the words chart at tomorrow's bell work.

Getting Started and 3.1 are starred. TE pacing is 3 days (11/11, 11/12, 11/16), matching the district calendar.